

Physically Debugging a Circuit

- We troubleshoot/debug an arduino circuit when it's not working as expected

Systematic Debugging Steps

1. Make sure your arduino has power / is plugged in, but make sure to unplug it from power while building your circuit
2. Check for open circuits (parts that are not electrically connected to each other)
 - On a breadboard, holes in the same row of a-e are electrically connected
 - All holes in the - column on the left side of the breadboard are connected
 - a-e is not connected with f-j, just as the + and - columns on the left are not connected with the + and - columns on the right
3. Make sure your jumper wires are connected to the correct arduino pins that are used in your code
4. Make sure that parts are oriented correctly
 - For parts like resistors, this doesn't matter
 - LEDs have polarity (+ side and - side), so orientation matters.
5. Electronic components can be broken, even if there is no sign of physical damage
 - For example, an LED can be burnt out if an incorrect resistor was used with it
 - Use a resistor calculator to check how many ohms a part resists
6. Less likely, but individual input/output pins could be damaged. Try switching to another pin and update your code to use that pin if you've eliminated all other possible issues
7. Do NOT connect the power bus (+ pin holes) directly to the ground bus (- pin holes) as this will create a short circuit between 5V and GND on the arduino, causing a large amount of current to flow and possibly damaging your circuit or arduino
8. If your circuit works inconsistently, check for loose connections

